

Academic Collaboration Network

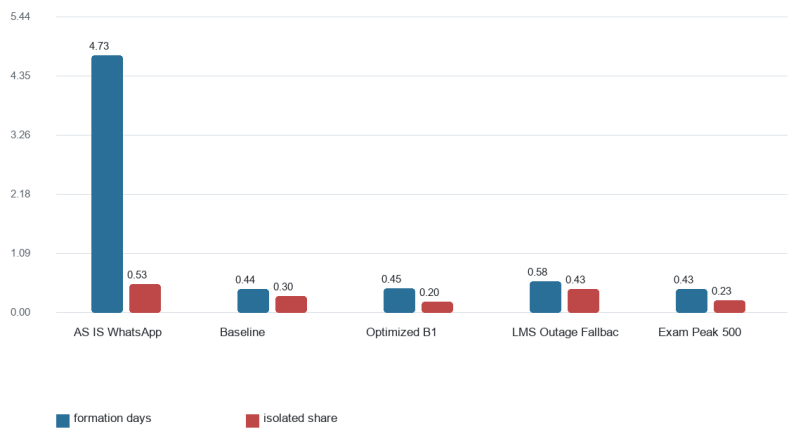
System Simulation and Validation Report

This report summarizes the simulation outputs generated by the final project code. The process-oriented model evaluates group formation workf

Process Simulation Summary

AS_IS_WhatsApp: success=0.47, days=4.73, latency=N/A, isolated=105.7
ACN_Baseline: success=0.96, days=0.44, latency=0.26s, isolated=61.0
ACN_Optimized_B1: success=0.98, days=0.45, latency=0.21s, isolated=39.4
ACN_LMS_Outage_Fallback: success=0.79, days=0.58, latency=0.31s, isolated=85.8
ACN_Exam_Peak_500: success=0.98, days=0.43, latency=1.26s, isolated=114.6

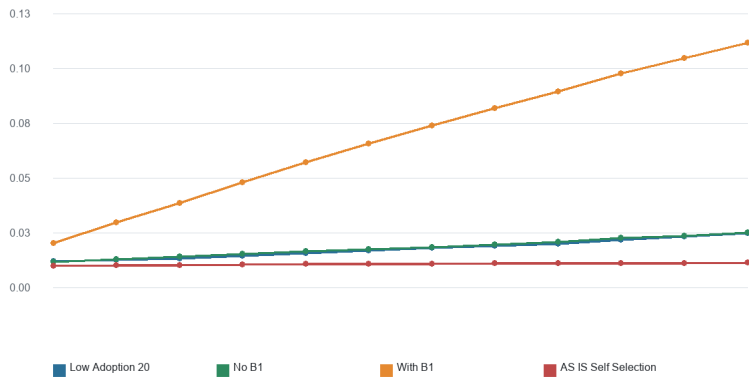
Process Simulation: Time and Isolation



Behavior Simulation Summary

AS_IS_Self_Selection: density=0.012, isolated=29, gini=0.40, satisfaction=0.45
ACN_No_B1: density=0.026, isolated=7, gini=0.43, satisfaction=0.63
ACN_With_B1: density=0.116, isolated=5, gini=0.19, satisfaction=0.79
ACN_Low_Adoption_20: density=0.026, isolated=11, gini=0.47, satisfaction=0.62

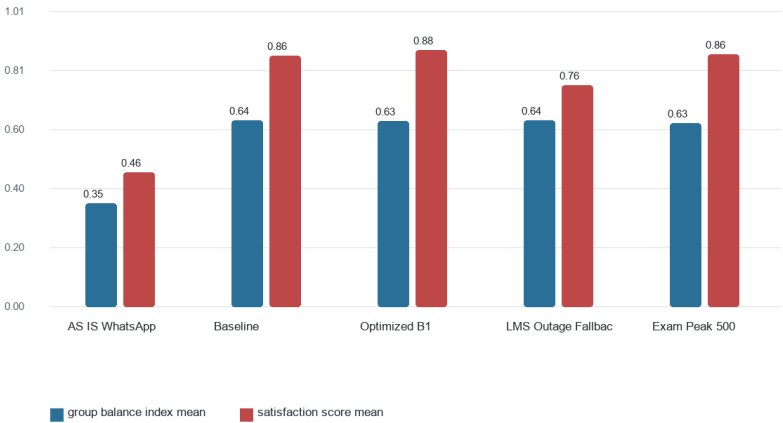
Behavior Simulation: Network Density



Validation Conclusions

The simulation validates the main ACN design decisions: the matching engine remains under the three-second target in the modeled stress scenarios.

Matching Quality and Satisfaction



Sensitivity: Remaining Isolated Students

